Solubility

1. The solubility of a gas in water depends on
   (a) Nature of the gas
   (b) Temperature
   (c) Pressure of the gas
   d) All of the above

2. Which of the following is not correct for D_2O
   (a) Boiling point is higher than H_2O
   (b) D_2O reacts slowly than H_2O
   (c) Viscosity is higher than H_2O at 25°
   (d) Solubility of *NaCl* in it is more than H_2O

3. The statement " The mass of a gas dissolved in a given mass of a solvent at any temperature is proportional to the pressure of the gas above the solvent" is
   (a) Dalton's Law of Partial Pressures
   (b) Law of Mass Action
   (c) Henry's Law
   (d) None of these

4. Which is correct about Henry's law
   (a) The gas in contact with the liquid should behave as an ideal gas
   (b) There should not be any chemical interaction between the gas and liquid
   (c) The pressure applied should be high
   (d) All of these

5. The statement "If 0.003 moles of a gas are dissolved in 900 *g* of water under a pressure of 1 atmosphere, 0.006 *moles* will be dissolved under a pressure of 2 atmospheres", illustrates
   (a) Dalton's law of partial pressure
   (b) Graham's law
   (c) Raoult's law
   (d) Henry's law

6. The solution of sugar in water contains
   (a) Free atoms
   b) Free ions
   (c) Free molecules
   d) Free atom and molecules